localEd: dependable document and speech workflows for a school, on one local machine

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Portfolio repository and runnable demo

The problem

Teachers at an IB international school in Korea run oral assessments (about 240 recordings a year are planned: ~180 English, ~60 Korean) and live in a sea of school documents: rubrics, guidelines, policies, forms, many of them Word, PDF or Hangul (HWP) files. Transcribing a recording, finding the right rubric, and writing feedback against it are slow, repetitive jobs. Cloud AI tools were not an option for student recordings and records, and teachers will not adopt a new app. **Goal:** a teacher types a request in Google Chat, and the work happens on a machine inside the school, with sources available for review and AI-suggested marks reviewed by a teacher.

What I did

I am a Computer Science and English teacher. I own this project end to end: I defined the workflow, set the privacy rules, chose the architecture, and decided what "done" means. I built it with AI coding agents (Claude Code and Codex) under my direction: I specified each change, reviewed and tested it, and deployed it on the school's Mac Studio. I also set up the evaluation: what to measure, on which data, and which claims are not yet allowed.

Architecture and key decisions

Workflow: Google Chat → local agent and retrieval → local models → Google Drive results. View the full architecture diagram.

- **Everything runs on-premise.** Whisper (MLX) for speech, a Qwen model via Ollama for reasoning and tool calling, bge-m3 for embeddings, sherpa-onnx for speaker diarization, and the OS's own OCR. Google Workspace is the only external service. Google Chat reaches the machine via Pub/Sub *pull*, so the school network needs no inbound ports.
- **Document ingestion for real school files:** Google Docs/Sheets/Slides, PDF with OCR fallback for scanned pages, DOCX/PPTX/XLSX/HWPX parsed with the standard library, legacy HWP/DOC/XLS/PPT, images and recordings. Text is chunked and indexed (SQLite FTS5 + vectors) for hybrid search.
- **Source and grading checks:** The code checks numbered citation identifiers. Quotes are matched approximately against source excerpts and unmatched quotes are flagged. Source support still needs teacher review. Marks are clamped to the rubric and graded three times (median kept). Every AI mark is labelled "AI-suggested, pending teacher review". The agent is re-prompted if it claims an action that no tool actually performed.
- **Access and privacy:** Drive folders are allow-listed (knowledge folders are read-only). Results are saved in a configured Drive folder and access is granted to the requester, subject to existing Drive permissions. Logs record counts, timings and error types, never student content. Recordings are deleted after decoding.

Measured results, with their exact scope

A benchmark on five English practice speaking assessments (24 min 15 s of audio, one Apple M4 Max, 10 Nov 2026) found:

- **All 5 of 5 pipeline runs completed, in 180.61 s total** (about 8× faster than real time). That covers Drive download, decoding, speech recognition, Q&A detection, diarization, name checks and rendering. Google Doc creation and Chat transport were not included.
- **Speech recognition took 39.75 s with Whisper large-v3-turbo**, against 78.23 s for full large-v3.
- **The test suite passes (225 tests)** on Linux; macOS-only parts are excluded there.

Not yet measured: accuracy against a human-checked transcript, speaker attribution, agreement with teachers' grades, Korean recordings, and teacher time saved. I don't report an accuracy figure until a teacher-verified reference set exists (details: docs/evidence.md).

What failed, and what I changed because of it

- **Speakers were mixed.** Teacher questions and student answers sometimes shared one speaker label. In one recording the two voices were never separated. Forcing "two speakers" made it worse, so a fix that keeps the count automatic and merges stray micro-speakers is drafted; it is not yet merged or re-benchmarked.
- **One recording got no Q&A boundary**, even though its own transcript contained the teacher's questions. It is the first regression test on the backlog.
- **The configured rubric was a placeholder for a different assessment** (/40 instead of /30). Grading was not run after review identified the placeholder, exposing a bigger issue: grading should never start from an unchecked rubric.
- **Speech recognition can change learner grammar.** Two transcripts differed between "they wearing" and "they are wearing". Listening is needed to establish what was said before using either version to assess language.

The next document-processing improvement

The placeholder-rubric failure points at a document problem. localEd's Word reader was built for search, and it flattens tables: a synthetic rubric becomes **58 loose paragraphs**, with band marks separated from their descriptors. The demo in this repository fixes that step:

1. **Retrieve** the rubric from allow-listed folders, citing file and position. A file outside the allow-list, including via ../ or a symlink, is refused.
2. **Extract** it table-aware into criteria, band marks and descriptors. Every value carries its source cell (e.g. *table 1, row 2, column 4*).
3. **Validate** with explicit rules: bands must increase, every band needs a descriptor, no duplicates, and the total must match the document. A broken draft yields three located errors.
4. **Require teacher approval** of the structured JSON (SHA-256 fingerprint) in the demo. Editing that JSON voids the approval; integrating the gate with localEd remains planned.

It is deterministic, uses only the standard library, and has 16 tests. Its output loads unchanged in localEd's grader format. **Next:** a small extraction benchmark (field-level exact match and source-reference correctness on synthetic and public rubrics and forms), then the same pattern for other school forms. In time, a layout-aware document parser could replace the hand-written reader where scans and complex layouts need it.